

100_Power_Regulation

200_FPGA_Power

300_FPGA_Config

400_DDR3L

500_FPGA_Banks_14_15

600_FPGA_Banks_34_35

700_Connectors

Layer	Material	Thickness (mm)	Thickness after lamination(mm)
L1-CU	Outer Base Copper 0.50Z	0.0175	0.0175 (Plating to 10Z)
PP	2116 RC58% DK 4.45	0.1300	0.1055
L2-CU	Inner Copper 10Z	0.0350	
CORE	Core DK 4.6	0.1300	0.2 (Core with Cu)
L3-CU	Inner Copper 10Z	0.0350	
PP	7626 RC46% DK 4.74	0.1960	0.1470
L4-CU	Inner Copper 10Z	0.0350	
CORE	Core DK 4.6	0.1300	0.2 (Core with Cu)
L5-CU	Inner Copper 10Z	0.0350	
PP	7626 RC46% DK 4.74	0.1960	0.1470
L6-CU	Inner Copper 10Z	0.0350	
CORE	Core DK 4.6	0.1300	0.2 (Core with Cu)
L7-CU	Inner Copper 10Z	0.0350	
PP	7626 RC46% DK 4.74	0.1960	0.1470
L8-CU	Inner Copper 10Z	0.0350	
CORE	Core DK 4.6	0.1300	0.2 (Core with Cu)
L9-CU	Inner Copper 10Z	0.0350	
PP	2116 RC58% DK 4.45	0.1300	0.1055
L10-CU	Outer Base Copper 0.50Z	0.0175	0.0175 (Plating to 10Z)

Conductor Impedance

Conductor Width (W)
0.1814 mm

Conductor Height (H1)
0.1055 mm

Conductor Height (H2)
0.13 mm

Frequency (MHz)
400

Formula Selection

Default

IPC-2141

Wadell

Wide calculation mode

Zo

50.0059 Ohms

Lo

2.9551 nH/cm

Co

1.1818 pF/cm

Tpd

59.0953 ps/cm

Conductor Impedance

Conductor Width (W)
0.117 mm

Conductor Height (H1)
0.1470 mm

Conductor Height (H2)
0.13 mm

Formula Selection

Default

IPC-2141

Wadell

Wide calculation mode

Zo

50.001 Ohms

Lo

3.6171 nH/cm

Co

1.4312 pF/cm

Tpd

71.9503 ps/cm

Differential Pairs

Conductor Width (W)
0.1086 mm

Conductor Spacing (S)
0.15 mm

Conductor Height (H)
0.1055 mm

Applied Voltage
0 Volts

Coupled Length
0 mm

Signal RiseTime
0 ns

Target Zdiff
95 Ohms

+/- Tolerance = 10%

Target Zdiff Plus
106.875 Ohms

Target Zdiff Minus
83.125 Ohms

Formula Restrictions:
0.1 < W/H < 3.0
0.1 < S/H < 3.0

Send to Via Calculator

Differential Protocol
DDR2 CLK/DQS

Zo
95.001 Ohms

Zodd
54.137 Ohms

Zeven
47.500 Ohms

61.701 Ohms

Kb (Term)
0.0653

Kb' (Unterm)
0.1300

-23.702

-17.718

0.0000 V

0.0000 V

Via Characteristics

IPC-2152 with modifiers mode

Via Hole Diameter
0.2 mm

Internal Pad Diameter
0.4 mm

Ref Plane Opening Diam
0.8 mm

Via Height
1.57 mm

Via Plating Thickness
0.0254 mm

Via Capacitance
0.4009 pF

Via Inductance
1.3963 nH

Via Impedance
59.016 Ohms

Via DC Resistance
0.00189 Ohms

Resonant Frequency
6726.824 MHz

Step Response
26.0257 ps

Power Dissipation
0.00577 Watts

Conductor Cross Section
0.0180 Sq.mm

Via Current
1.7467 Amps

Via Characteristics

Differential Pair Z

Baud Rate
1 Gbaud

Anisotropy of Material
18 %

Maximum Stub Length
8.7493 mm

Insertion Loss
-0.0257 dB

Drill Hole Diameter
0.2 mm

Full Plane Opening

Ref Plane Opening H
1.3 mm

Ref Plane Opening W
0.8 mm

Via Spacing S
0.7 mm

Odd Mode Impedance
47.360 Ohms

Differential Impedance
94.719 Ohms

Er Effective
5.947

Differential Pairs

Conductor Width (W)
0.1 mm

Conductor Spacing (S)
0.21 mm

Conductor Height (H1)
0.1470 mm

Conductor Height (H2)
0.13 mm

Applied Voltage
0 Volts

Coupled Length
0 mm

Signal RiseTime
0 ns

Target Zdiff
100 Ohms

+/- Tolerance = 10%

Target Zdiff Plus
112.500 Ohms

Target Zdiff Minus
87.500 Ohms

Formula Restrictions:
0.1 < W/H < 3.0
0.1 < S/H < 3.0

Send to Via Calculator

Differential Protocol
DDR3 CLK/DQS

Zo
95.463 Ohms

Zodd
50.409 Ohms

Zeven
47.732 Ohms

53.236 Ohms

Kb (Term)
0.0273

Kb' (Unterm)
0.0545

-31.284

-25.269

0.0000 V

0.0000 V

Project References:

<https://digilent.com/reference/programmable-logic/arty-a7/reference-manual>

https://digilent.com/reference/_media/programmable-logic/arty-a7/arty-a7-e2-sch.pdf

<https://github.com/snhobbs/scopefun-hardware/tree/master/KiCadSource>

<https://github.com/arm8686/FT2232HL-Board/blob/master/FT2232HLx20Boardx20schx20V3.6.pdf>

AMD Documentation:

https://docs.amd.com/v/u/en-US/ug470_7Series_Config

https://docs.amd.com/v/u/en-US/ds181_Artix_7_Data_Sheet

<https://docs.amd.com/v/u/en-US/wp484-a7-s7-ddr2-3-pcb>

https://docs.amd.com/r/en-US/ug908-vivado-programming-debugging/JTAG-Cables-and-Devices-Supported-by-hw_server

https://docs.amd.com/r/en-US/ug586_7Series_MIS/Trace-Lengths

High-speed design:

Project Goals:

Run ML/MPC/LQR on an FPGA

Use DDR3 to load weights/matrices from an SD card

Stackup:

L1: High Speed

L2: Ground

L3: Power

L4: Power

L5: Ground

L6: High Speed

L7: Ground

L8: High Speed

L9: Ground

L10: High Speed

DDR Routing:

According to WP484, route single ended as 50ohm, diff as 100ohm

Chose closest ODT to 50ohm

Note: Due to stackup, diff pairs will be routed as 95ohm

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Sheet: /

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Title: FPGA_AURA

Size: A4

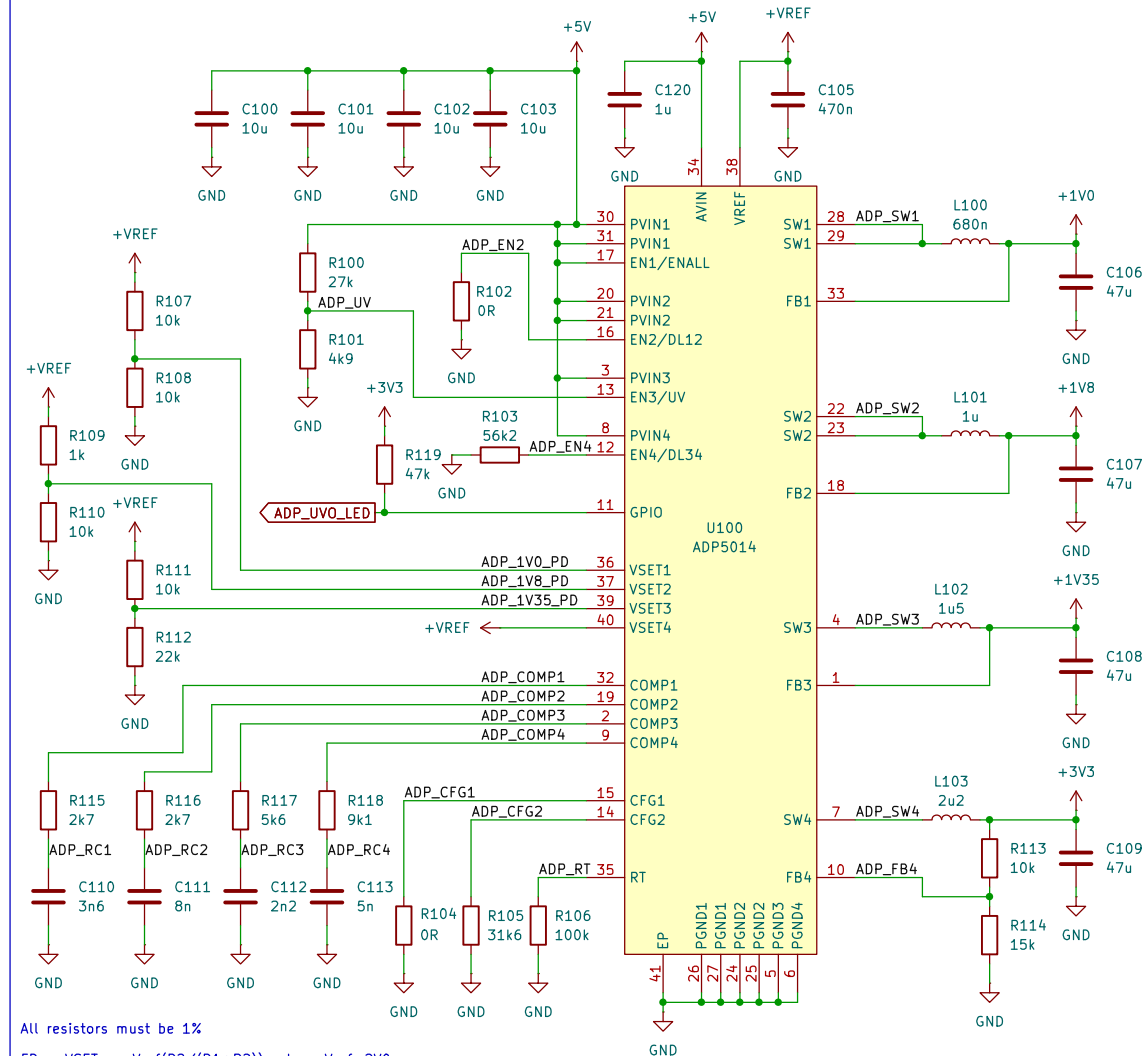
Date: 13/02/2026

KiCad E.D.A. 10.0.1

Rev: V0.1

Id: 1/8

Quad Buck Regulator (1V0/4A, 1V35/4A, 1V8/2A, 3V3/2A)



All resistors must be 1%

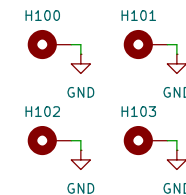
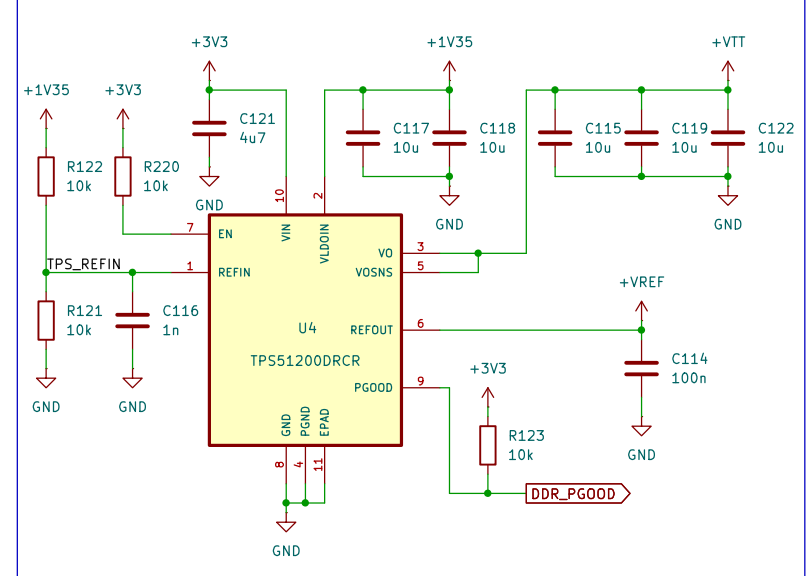
FB = VSETx = Vref(R2/(R1+R2)), where Vref=2V0
For 3V3, tie Vset to 2V0 and use PD on FB
 $L = (V_{in} - V_{out})(V_{out}/V_{in}) / (0.3I * f_{sw})$

Compensation network assumes maximum current draw on each channel
due to the unpredictable FPGA current draw

Power Sequence 1V0 @ 0ms, 1V8 @ 6ms, 1V35 @ 12ms, 3V3 @ 12ms

RT: 1MHz
CFG1: 4A, 4A, 2A, 2A
CFG2: Sequence: 1x, FPWM, UV0
EN3/UV: 3V9 cut-off
EN2/DL12: 2ms (soft start), 6ms
EN4/DL34: 12ms, 12ms

DDR3L VTT Regulator



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Sheet: /100_Power_Regulation/
File: Power_Regulation.kicad_sch

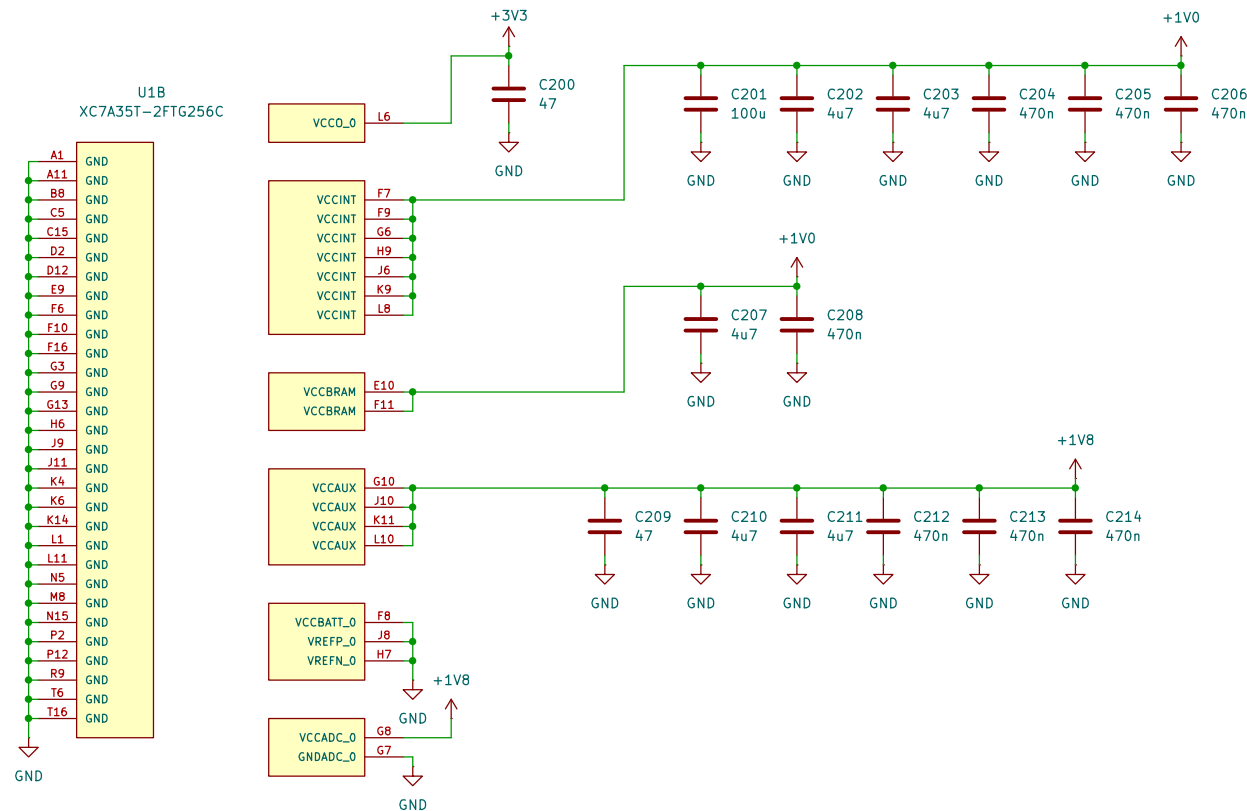
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Size: A4 Date: 01-11-2025

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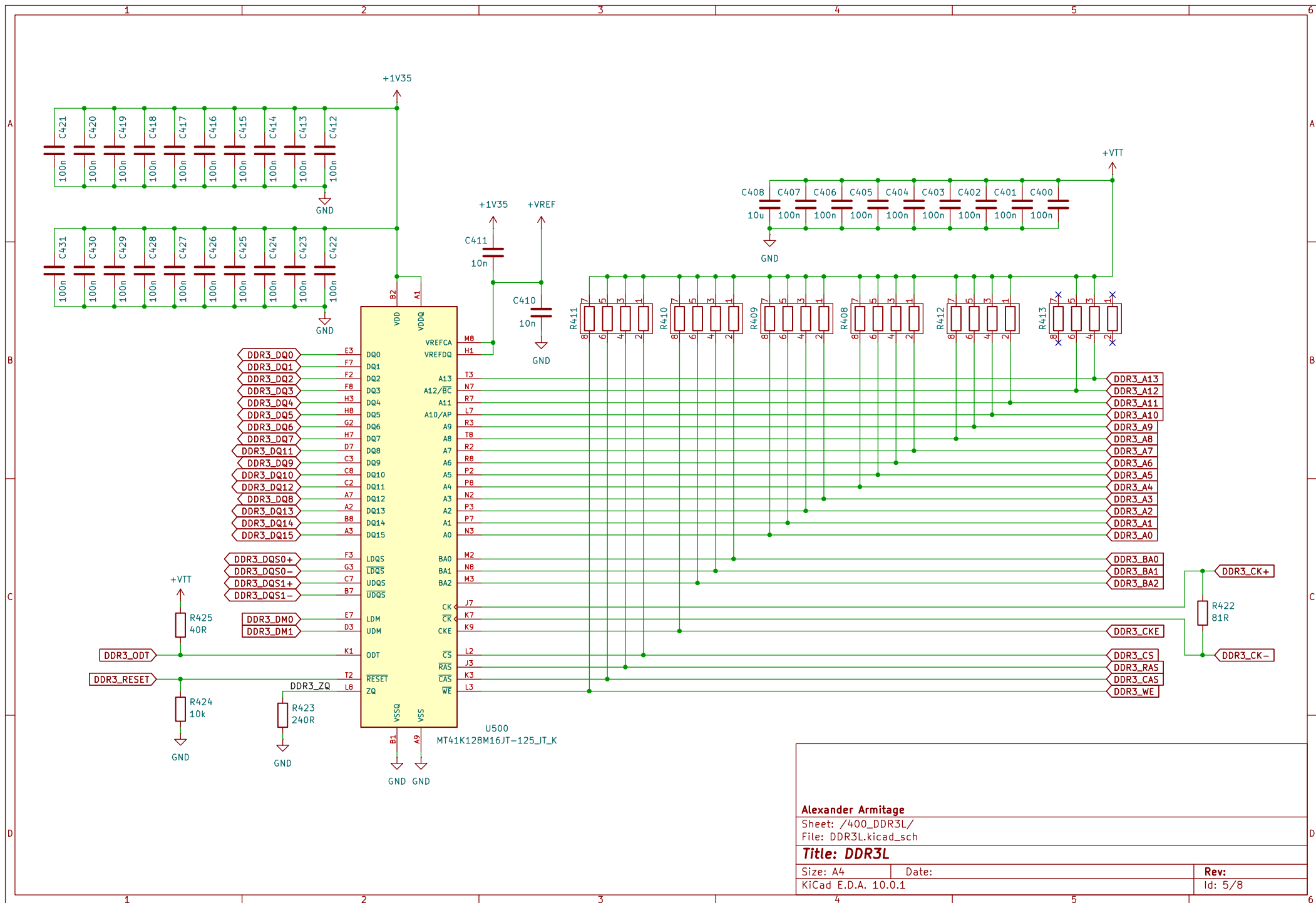
Rev: V0.1

Id: 2/8



Power on sequency VCCINT -> VCCBRAM -> VCCAUX -> VCCO.
If VCCAUX and VCCO are the same, they can be powered on simultaneously.

Id: 4/8



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Sheet: /400_DDR3L/

File: DDR3L.kicad_sch

Title: DDR3L

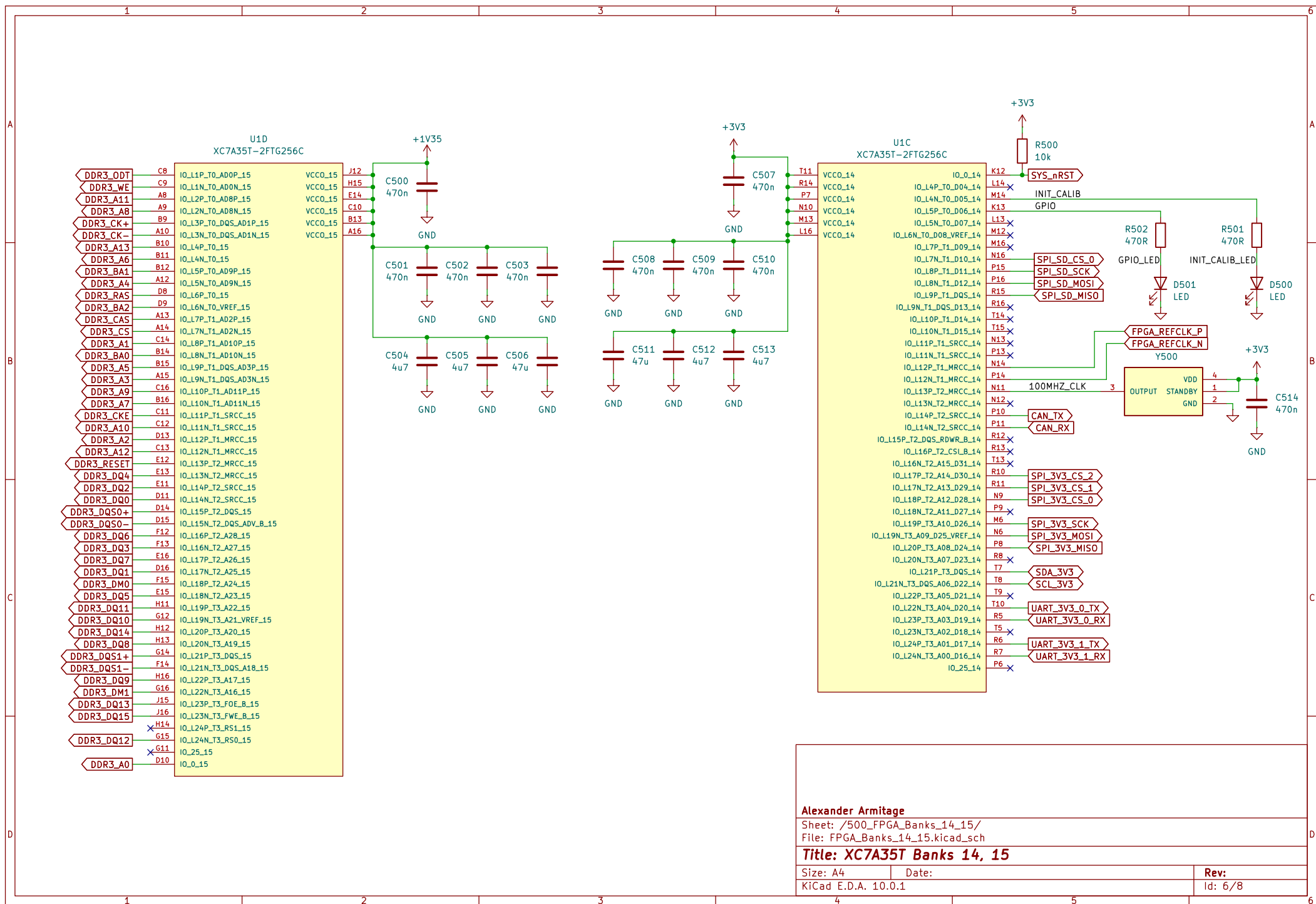
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Date:

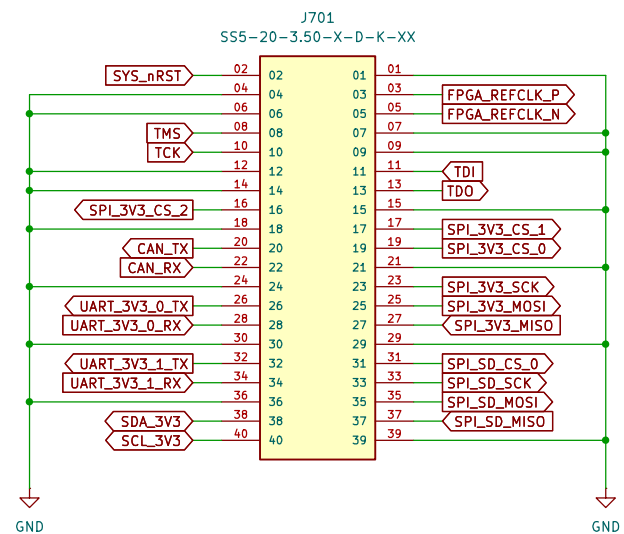
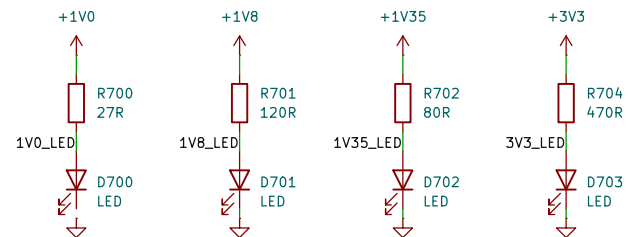
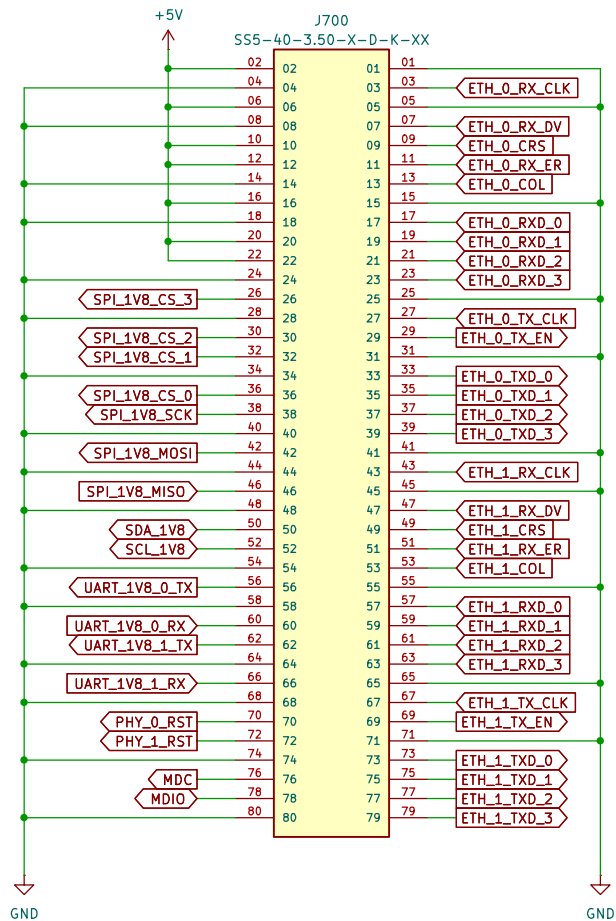
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Rev:

Id: 5/8







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Sheet: /700_Connectors/
File: Connectors.kicad_sch

Title: Connectors

Size: A4
KiCad E.D.A. 10.0.1

Date:
Rev: 8/8